

Basketball

Note: For the purposes of this content, the basketball programming will be a methodology shared across “big” (centers and forwards) and “small” (guards). Through years of clinical application of the OPT™ Model with basketball players, from youth leagues through the professional association, there has not been a need to treat the athletes differently from a strength and conditioning aspect. At a fundamental level, all positions on the court experience the same movement patterns and energy expenditure demands; therefore, we will focus on how the OPT Model can be used for all basketball positions alike. If the Sports Performance Coach needs to modify the model based on the athlete’s needs, then it is very appropriate to do so based on the assessment process and the goals set for the athlete.

General Information (McClay et al., 1994a, 1994b)

Demands of the Sport	
Jumps classification	• Low: A shot or unchallenged rebound • Medium: Most rebounds, defending jump shots, and a jump shot • High: Maximal or near maximal as would occur with a dunk, blocked shot, or challenged jump shot
Jumps in a game	• Low: 30% • Medium: 45% • High: 25%
Average number of jumps per game: 70	• Guards: 55 • Centers: 83 • Forward: 72
Ground reaction forces	• Running: ○ Vertical: $3 \times$ body weight (BW) ○ Anterior/posterior: $0.5 \times$ BW ○ Mediolateral: $0.25 \times$ BW • Vertical forces: ○ Starting: $0.8 \times$ BW ○ Lay-up landing: $8.9 \times$ BW ○ Stopping: $2.7 \times$ BW ○ Cutting: $3.0 \times$ BW • Anterior/posterior forces: ○ Stopping: $1.3 \times$ BW • Mediolateral direction: ○ Shuffling: $1.4 \times$ BW

Assessment & Program Strategies: Basketball

Activity Information (*Gambetta, 1999*)

Demands of the Sport	
Total game time (professional): 48 minutes	• 4 × 12-minute quarters • 15-minute halftime • 2 minutes between quarters
Activity type	• Walking or standing: 4 minutes • Jogging: 4 minutes • Running: 4 minutes • Sprinting: 3 minutes • Shuffling (low to medium intensity): 9 minutes • Shuffling (high intensity): 2 minutes • Jumping: 41 seconds
Movement	• While the ball is in play, a change in movement category happens every 2 seconds • There are 1,000 different movements during a game • 28% of all court time is spent in strenuous exertion • Intense activity occurs for 13–14 seconds at a time • 105 intense activities per game, every 21 seconds • 31% of the game is devoted to side-to-side movements (66% of these movements are intense efforts) • Individual shuffle movements are 1–4 seconds in duration • Sprints are 1–5 seconds in duration

Suggested Assessments for Basketball

Assessment Type	Assessment Method
Transitional postural assessments	• Overhead squat test • Single-leg squat test • Split squat • Star excursion balance test
Mobility assessments (if needed)	• Foot and ankle testing • Lumbo-pelvic-hip complex testing • Shoulder and spine testing
Dynamic movement assessments	• Drop jump • Shark skill test • Davies test • Triple crossover hop test
Aerobic test (choose one of the following)	• Beep test • Yo-yo intermittent recovery test
Anaerobic test	• 300-yard shuttle run
Muscular endurance assessment	• Push-up test • Pull-up test • Curl-up test
Muscular strength assessment	• 1–5 RM bench press • 1–5 RM squat • Nordic hamstring lower test
Power assessments	• Vertical jump test ◦ Double-leg ◦ Single-leg ◦ 15-foot approach (single- or two-leg) • Horizontal broad jump (single-leg) • Medicine ball soccer throw • Rotational medicine ball throw test (left and right)
Speed assessments (Norms may not exist for the following tests. Use these as objective measures of improvement over time.)	• 40-yard sprint test • $\frac{3}{4}$ court sprint (baseline to farthest from the throw line)
Agility assessments	• LEFT (lower-extremity functional) test • 5–10–5 (Pro agility test) • 5–0–5 test

Off-Season Goals and Considerations for Basketball

Goals	Considerations
• Physical and mental reconditioning • Correcting impaired movement patterns based on results of transitional postural assessments • Improving range of motion, joint stability, and muscle strength and endurance ○ Typical areas of focus for enhanced flexibility or range of motion include ankle dorsiflexion, hip internal rotation, rectus femoris, and lats. ○ Typical areas of focus for enhanced strengthening include gluteus medius, pectineus, gluteus maximus, posterior tibialis, medial gastrocnemius, and core musculature. • Improving: ○ Cardiorespiratory endurance (aerobic/anaerobic) ○ Total-body stabilization, strength, and power • Increasing reaction time, speed, agility, and quickness	• Planning and programming will be based on assessment, client goals, and the demand of the sport. • A systematic off-season plan will incorporate integrated training, including: ○ Core training ○ Balance training ○ Plyometric training ○ Speed, agility, and quickness training ○ Integrated resistance training ○ Cardiorespiratory training combined with a proper nutrition plan to maximize training and recovery • An off-season training schedule will be developed by working backward from the commencement of the preseason to allow for maximum injury prevention and performance gains without the athlete overtraining. • Opportunities for appropriate rest will be provided between workouts. • Considerations will include: ○ Basketball conditioning will be as like the game as possible. ○ Although the game of basketball is a very dynamic activity, many basketball players enter the off-season having diminished stabilization mechanisms that must be trained and reconditioned. • Performance programs will be modified appropriately, from a volume and exercise standpoint, to accommodate basketball-specific activities to avoid overtraining and maximize results. The athlete will continue to increase basketball skill activities as the off-season progresses.

Assessment & Program Strategies: Basketball



Sample Off-Season OPT Basketball Plan (May–September)

Level	Phase	JAN	FEB	MAR– APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV– DEC
		In- Season	In- Season	In- Season	Off- Season (1)	Off- Season (2)	Off- Season (3)	Off- Season (3)	Off- Season (5)	Pre- Season	In- Season
Corrective					X						
Stabilization	1					X			X		
Strength	2						X		X		
	3										
	4										
Power	5		X					X	X		
	6										

Sample Off-Season Weekly OPT Basketball Schedule

Month 1: 3 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Corrective	OFF	Corrective	OFF	Corrective	OFF	OFF

Month 2: 3 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Stabilization Endurance	OFF	Stabilization Endurance	OFF	Stabilization Endurance	OFF	OFF
Other	On-court conditioning				On-court conditioning		

Month 3: 5 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Stabilization Endurance Total body	OFF	Stabilization Endurance Upper body	OFF	Stabilization Endurance Lower body	OFF	OFF
Other	On-court conditioning			On-court conditioning			

Month 4: 5 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Power	OFF	Power	OFF	Power	OFF	OFF
Other	On-court conditioning	On-court conditioning		On-court conditioning			

Month 5: 5 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Strength Endurance		Stabilization Endurance		Power	OFF	OFF
Other	On-court conditioning			On-court conditioning			

Sample Off-Season OPT Basketball Programs

Corrective Exercise Training

INHIBIT				Sets	Reps	Time
Myofascial Rolling						
Calves				1		30 sec
IT band				1		30 sec
Tensor fascia latae				1		30 sec
LENGTHEN				Sets	Reps	Time
Gastrocnemius stretch				1		30 sec
Soleus				1		30 sec
Supine biceps femoris stretch				1		30 sec
ACTIVATE		Sets	Reps	Tempo	Intensity	Rest
Posterior tibialis strengthening		2	12	4/2/1		0 sec
Single-leg calf raise		2	12	4/2/1		0 sec
Single-leg hip adduction: Pectineus		2	12	4/2/1		0 sec
Single-leg hip abduction: Gluteus medius		2	12	4/2/1		90 sec
INTEGRATE		Sets	Reps	Tempo	Intensity	Rest
Total body	Step-up, balance to overhead press	2	12	4/2/1	60%	90 sec

Assessment & Program Strategies: Basketball

Phase I: Stabilization Endurance Training

WARM-UP			Sets	Reps	Time	
Myofascial Rolling						
Tensor fascia latae			1		30 sec	
Piriformis			1		30 sec	
Static Stretching						
Gastrocnemius stretch			1		30 sec	
Kneeling hip flexor stretch			1		30 sec	
Standing biceps femoris stretch			1		30 sec	
Dynamic Stretching		Sets	Reps	Tempo		
Lunge with rotation		1	10	Moderate	0 sec	
Medicine ball woodchop		1	10	Moderate	0 sec	
CORE AND BALANCE		Sets	Reps	Tempo	Rest	
Side iso-ab		2	15	4/2/1	0 sec	
Floor bridge		2	15	4/2/1	0 sec	
Single-leg balance reach		2	15	4/2/1	60 sec	
PLYOMETRICS		Sets	Reps	Tempo	Rest	
Box jump-down with stabilization		2	8	Hold landing	60 sec	
SPEED/AGILITY/QUICKNESS		Sets	Reps	Tempo	Rest	
Speed ladder: Six exercises		2			60 sec	
RESISTANCE		Sets	Reps	Tempo	Intensity	Rest
Total body	Step-up, balance to overhead press: Sagittal plane	2	15	4/2/1	65%	0 sec
Chest	Cable chest press	2	15	4/2/1	65%	0 sec
Back	Single-leg cable row	2	15	4/2/1	65%	0 sec
Shoulders	Single-leg lateral raise	2	15	4/2/1	65%	0 sec
Legs	Single-leg Romanian deadlift	2	15	4/2/1	65%	90 sec
COOL-DOWN						
Myofascial rolling and static stretching						

Assessment & Program Strategies: Basketball

Phase 2: Strength Endurance Training

WARM-UP		Sets	Reps	Time
Myofascial Rolling				
IT band		1		30 sec
Piriformis		1		30 sec
Calf		1		30 sec
Active Stretching				
Standing adductor stretch		1	10	
Standing hip flexor stretch		1	10	
Floor lat stretch		1	10	
Dynamic Stretching		Sets	Reps	Tempo
Multiplanar lunge		1	10	Moderately fast
Squat to overhead reach		1	10	Moderately fast
CORE AND BALANCE		Sets	Reps	Tempo
Knee-ups		2	12	1/1/1
Back extension		2	12	1/1/1
Single-leg squat touchdown		2	12	1/1/1
PLYOMETRICS		Sets	Reps	Tempo
Power step-ups		2	8	Repeating
Butt kicks		2	8	Repeating
SPEED/AGILITY/QUICKNESS		Sets	Reps	Tempo
Speed ladder: Six exercises		2		
Box drill		6		
LEFT test		2		
RESISTANCE		Sets	Reps	Tempo
Total body	Optional			
Chest	Machine press	3	8	2/0/2
	Cable chest press		12	4/2/1
Back	Seated lat pull-down	3	8	2/0/2
	Standing cable row: alternating arm		12	4/2/1
Shoulders	Standing lateral raise	3	8	2/0/2
	SL DB press		12	4/2/1
Legs	Barbell squats	3	8	2/0/2
	Side lunge to balance		12	4/2/1
COOL-DOWN				
Myofascial rolling and static stretching				

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WARM-UP			Sets	Reps	Time	
Myofascial Rolling						
Calf			1		30 sec	
Tensor fascia latae			1		30 sec	
Piriformis			1		30 sec	
Dynamic Warm-up (Some athletes may still require static stretching prior to the dynamic warm-up.)						
Lateral tube walking			1	10	0 sec	
Prisoner squats			1	10	0 sec	
Multiplanar lunge			1	10	0 sec	
Leg swings: Front to back			1	10	0 sec	
Leg swings: Side to side			1	10	0 sec	
MB rotations			1	10	0 sec	
Russian twist			1	10	0 sec	
Multiplanar hop to stabilization			1	10	60 sec	
CORE AND BALANCE		Sets	Reps	Tempo	Rest	
No CORE AND BALANCE (incorporated into dynamic warm-up)						
PLYOMETRICS		Sets	Reps	Tempo	Rest	
No PLYOMETRICS (incorporated into dynamic warm-up))						
SPEED/AGILITY/QUICKNESS		Sets	Reps	Tempo	Rest	
Speed ladder: Eight exercises		1			60 sec	
Box drill		4			15 sec	
T-drill		4			25 sec	
Shuttle: Six lengths of basketball court		2			60 sec	
RESISTANCE		Sets	Reps	Tempo	Intensity	Rest
Total body	DB squat, curl to overhead press	3	5	Controlled	85%	0 sec
	DB snatch		10	Explosive	45%	2 min
Chest	DB chest press	3	5	Controlled	85%	0 sec
	Cable chest press		10	Explosive	2% of BW	2 min
Back	Seated lat pull-down	3	5	Controlled	85%	0 sec
	Standing cable row: alternating arm		10	Explosive	2% of BW	2 min
Shoulders	Incorporated in total-body exercises					
Legs	Barbell squat	3	5	Controlled	85%	0 sec
	Squat jump		10	Explosive of BW		3 min
COOL-DOWN						
Myofascial rolling and static stretching						

Preseason Goals and Considerations for Basketball

Goals	Considerations
- Correcting impaired movement patterns based on results of preseason assessments - Maintaining: - Physical health (decreases chance of injury for participation in practice and games, and optimal performance within such activities) - Range of motion, joint stability, and muscle strength and endurance - Cardiorespiratory endurance (aerobic/anaerobic) - Total-body stabilization, strength, and power - Speed, agility, and quickness	- A systematic preseason plan that transitions the athlete from performance enhancement to injury prevention and maintenance of off-season gains will be developed. This transition should be implemented with the consideration that the athlete's sport participation level is increasing dramatically from what they have become used to in the off-season. - Proper recovery and conditioning should be applied during this time of increased activity will be provided. - Consider the exercise volume of injury prevention and performance training based on the athlete's increased activity level.

Sample Preseason OPT Basketball Plan (October)

Level	Phase	JAN	FEB	MAR	APR	MAY-SEP	OCT	NOV	DEC
		In-Season	In-Season	In-Season	In-Season	Off-Season	Pre-Season (1)	In-Season	In-Season
Corrective							X		
Stabilization	1						X		
Strength	2						X		
	3								
	4								
Power	5								
	6								

Sample Preseason Weekly OPT Basketball Schedule

Month 1

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Strength Endurance Upper body	OFF	Corrective	OFF	Stabilization Endurance Lower body	OFF	OFF
Other		Game			Game		

Sample Preseason OPT Basketball Programs

Corrective Exercise Training

INHIBIT			Sets	Reps	Time	
Myofascial Rolling						
Tensor fascia latae			1		30 sec	
IT band			1		30 sec	
Adductors			1		30 sec	
LENGTHEN			Sets	Reps	Time	
Gastrocnemius stretch			1		30 sec	
Supine biceps femoris stretch			1		30 sec	
Standing adductor stretch			1		30 sec	
Kneeling hip flexor stretch			1		30 sec	
ACTIVATE		Sets	Reps	Tempo	Intensity	Rest
Side-lying hip abduction		2	10	4/2/1		0 sec
Suspension bridge		2	10	4/2/1		90 sec
INTEGRATE		Sets	Reps	Tempo	Intensity	Rest
Single-leg squat		2	12	4/2/1	60%	90 sec

Assessment & Program Strategies: Basketball

Phase I: Stabilization Endurance Training (Lower Body)

WARM-UP				Sets	Reps	Time	
Myofascial Rolling							
Tensor fascia latae				1		30 sec	
Piriformis				1		30 sec	
Thoracic spine				1		30 sec	
Static Stretching							
Gastrocnemius stretch				1		30 sec	
Standing adductor stretch				1		30 sec	
Standing biceps femoris stretch				1		30 sec	
Dynamic Stretching			Sets	Reps	Tempo		
Lunge with rotation			1	10	Moderate	0 sec	
Medicine ball woodchop			1	10	Moderate	0 sec	
CORE AND BALANCE			Sets	Reps	Tempo	Rest	
Side iso-ab			2	15	4/2/1	0 sec	
Floor bridge: Single-leg			2	15	4/2/1	0 sec	
Single-leg balance reach			2	15	4/2/1	60 sec	
PLYOMETRICS			Sets	Reps	Tempo	Rest	
Squat jump with stabilization			2	10	Hold landing	60 sec	
SPEED/AGILITY/QUICKNESS			Sets	Reps	Distance	Rest	
No SAQ because of on-court drills and conditioning							
RESISTANCE			Sets	Reps	Tempo	Intensity	Rest
Total body	Step-up, balance to overhead press: SA		2	15	4/2/1	65%	0 sec
Legs	Single-leg Romanian deadlift		2	15	4/2/1	65%	0 sec
	Single-leg calf raise						
	Lunge to balance: Frontal plane		2	15	4/2/1	65%	90 sec
COOL-DOWN							
Myofascial rolling and static stretching							

Assessment & Program Strategies: Basketball

Phase 2: Strength Endurance Training (Upper Body)

WARM-UP		Sets	Reps	Time		
Myofascial Rolling						
Thoracic spine		1		30 sec		
Lats		1		30 sec		
Active Stretching						
Floor lat stretch		1	10			
Standing pec stretch		1	10			
Dynamic Stretching		Sets	Reps	Tempo	Rest	
Rotation med ball throw		1	10	Moderately fast	0 sec	
Med ball back extension throw		1	10	Moderately fast	0 sec	
CORE AND BALANCE		Sets	Reps	Tempo	Rest	
No CORE AND BALANCE due to lack of time (core/balance type exercises incorporated in resistance training)						
PLYOMETRICS		Sets	Reps	Tempo	Rest	
No PLYOMETRICS because of increased activity						
SPEED/AGILITY/QUICKNESS		Sets	Reps	Distance	Rest	
No SAQ because of on-court drills and conditioning						
RESISTANCE		Sets	Reps	Tempo	Intensity	Rest
Chest	Bench press	3	12	2/0/2	70%	0 sec
	Push-up: Feet in suspension		8	4/2/1		
Back	Seated cable row	3	12	2/0/2	70%	0 sec
	SL cable pull-down		8	4/2/1		
Shoulders	Standing overhead press	3	12	2/0/2	70%	0 sec
	SL DB scaption		8	4/2/1		
COOL-DOWN						
Myofascial rolling and static stretching						

In-Season Goals and Considerations for Basketball

Goals	Considerations
• Maintaining: ○ Range of motion, joint stability, and muscle strength and endurance ○ Cardiorespiratory endurance (aerobic/anaerobic) ○ Total-body stabilization, strength, and power ○ Speed, agility, and quickness (position players) ○ Arm and shoulder strength • Establishing with the athlete a communication process for monitoring in-season maintenance, and a plan that works based on what the athlete needs, wants, and will have the capability to do while playing basketball	• Assessments and testing to ensure program considerations are being made to prevent overuse injuries and maintain performance levels, will be performed. • Impaired movement patterns with exercise strategies being implemented will be attended to. • Additional conditioning, for athletes who may not get the same type of on-the-court minutes during games as others, will be considered. • Athletes, who may get limited to no play time during games, will be considered for progression to the strength and power phases of the OPT Model to offset the lack of on-court power/explosive movements that they are missing due to lack of playing time. • Coaching styles, and their practice routines, will be assessed, and workouts will be adjusted accordingly in-season. Adjustments to avoid overtraining and undertraining are important. • A systematic in-season maintenance plan will be developed that allows the athlete to minimize decreases in performance and chance of injury over a long season with repetitive sport movement patterns. • The exercise volume of injury prevention, and performance training based on the athlete's increased travel and activity level, will be considered.

In-Season OPT Basketball Plan (November–April)

Level	Phase	JAN	FEB	MAR	APR	MAY–SEP	OCT	NOV	DEC
		In-Season (3)	In-Season (4)	In-Season (5)	In-Season (6)	Off-Season	Pre-Season	In-Season (1)	In-Season (2)
Corrective		X	X	X	X			X	X
Stabilization	1	X	X	X	X			X	X
Strength	2	X	X	X	X			X	X
	3								
	4								
Power	5								
	6								

Assessment & Program Strategies: Basketball



Sample In-Season Weekly OPT Basketball Schedule

Any Month

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Strength Endurance Upper body	OFF	Corrective	Corrective	Stabilization Endurance	OFF	Corrective
Other	Game			Game		Game	

Sample In-Season OPT Basketball Program

Corrective Exercise Training

INHIBIT			Sets	Reps	Time	
Myofascial Rolling						
Tensor fascia latae			1		30 sec	
Piriformis			1		30 sec	
Lats			1		30 sec	
LENGTHEN			Sets	Reps	Time	
Gastrocnemius stretch			1		30 sec	
Standing biceps femoris stretch			1		30 sec	
Kneeling hip flexor stretch			1		30 sec	
Floor lat stretch			1		30 sec	
ACTIVATE		Sets	Reps	Tempo	Intensity	Rest
Single-leg calf raise		1	15	4/2/1		0 sec
Single-leg hip extension		1	15	4/2/1		0 sec
Suspension bridge		1	15	4/2/1		90 sec
INTEGRATE		Sets	Reps	Tempo	Intensity	Rest
Total-body cable squat to row		3	10	4/2/1	60%	90 sec

Assessment & Program Strategies: Basketball

Phase I: Stabilization Endurance Training

WARM-UP			Sets	Reps	Time	
Myofascial Rolling						
Tensor fascia latae			1		30 sec	
Piriformis			1		30 sec	
IT band			1		30 sec	
Static Stretching						
Gastrocnemius stretch			1		30 sec	
Standing biceps femoris stretch			1		30 sec	
Kneeling hip flexor stretch			1		30 sec	
Supine piriformis stretch			1		30 sec	
Dynamic Stretching		Sets	Reps	Tempo		
Lunge with rotation		1	10	Moderate	0 sec	
Prisoner squats to calf raise		1	10	Moderate	0 sec	
CORE AND BALANCE		Sets	Reps	Tempo	Rest	
Side iso-abs		3	12	4/2/1	0 sec	
Floor cobra		3	12	4/2/1	0 sec	
Single-leg lift and chop		3	12	4/2/1	60 sec	
PLYOMETRICS		Sets	Reps	Tempo	Rest	
No PLYOMETRICS because of increased activity						
SPEED/AGILITY/QUICKNESS						
No SAQ because of on-court drills and conditioning						
RESISTANCE		Sets	Reps	Tempo	Intensity	Rest
Total body	Cable squat to row	3	12	4/2/1	70%	0 sec
Chest	Suspension SL chest press	3	12	4/2/1	70%	0 sec
Back	Incorporated in total-body exercise					
Shoulders	Single-leg overhead DB press	3	12	4/2/1	70%	0 sec
Legs	Lunge to balance: Frontal plane	3	12	4/2/1	70%	60 sec
COOL-DOWN						
Myofascial rolling and static stretching						

Assessment & Program Strategies: Basketball

Phase 2: Strength Endurance Training (Upper Body)

WARM-UP				Sets	Reps	Time	
Myofascial Rolling							
Piriformis				1		30 sec	
Adductor				1		30 sec	
Thoracic spine				1		30 sec	
Active Stretching							
Gastrocnemius stretch				1	10		
Standing adductor stretch				1	10		
Standing pec stretch				1	10		
Dynamic Stretching			Sets	Reps	Tempo	Rest	
Multiplanar lunge with reach			1	10	Moderately fast	0 sec	
Modified push up to rotation			1	10	Moderately fast	0 sec	
CORE AND BALANCE							
Cable chop			3	10	1/1/1	0 sec	
Single-leg squat			3	10	1/1/1	60 sec	
PLYOMETRICS			Sets	Reps	Tempo	Rest	
No PLYOMETRICS because of increased activity							
SPEED/AGILITY/QUICKNESS			Sets	Reps	Distance	Rest	
No SAQ because of on-court drills and conditioning							
RESISTANCE			Sets	Reps	Tempo	Intensity	Rest
Chest	Incline DB press		2	10	2/0/2	75%	0 sec
	Cable chest press			10	4/2/1	60%	
Back	Seated cable row		2	10	2/0/2	75%	0 sec
	SL lat pull-down			10	4/2/1	60%	
Shoulders	Seated DB scaption		2	10	2/0/2	75%	0 sec
	SL SA DB press			10	4/2/1	60%	
COOL-DOWN							
Myofascial rolling and static stretching							